DIP-CSE340 ASSIGNMENT 2
Deadline : 18/9/2025

Total Marks : 40 Marks | **Total Days : 14 Days**

Instructions:

- Standard institute plagiarism policy holds.
- This is an individual assignment. State any assumptions you have made clearly.
- Even a second late will be considered late, so do submit your assignments at least before 11:56 PM on the date of the deadline to avoid any unexpected issues. Late submissions are not entertained.
- Viva 5 marks will be taken during the demos (written/oral).
- Submit a Jupiter notebook (dip_rollnumber.ipny) with results from section A, B and C. You can use either Python/Matlab. Jupiter Notebook for Matlab is also available. Wherever required use text cell/comment to answer the questions. Feel free to submit a pdf (dip_rollnumber.pdf) with the theoretical answers.

Section I - Transformations

(CO1&2)(6.5 marks)

1. Consider the images in the Figure 1.

   (A) Compare the (RGB) histogram of the original image with histograms of the
   - log transformation : $c = 0.5, 2, 5$ and 10.
   - gamma transformation : $c = 0.5, 2, 5$ and 10; $\gamma = 0.5$ and 2

   (B) Perform histogram matching on the original image (b) (source) using the gamma transformed images of (c)(reference).

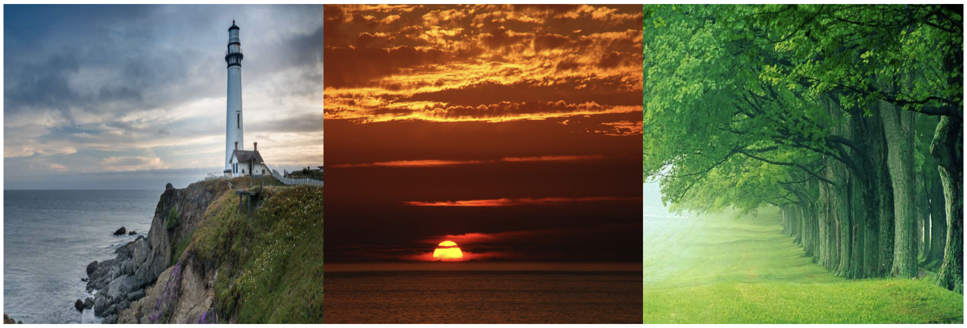
   (C) Perform histogram matching on the original image (a) (source) using the log transformed images of (c)(reference).

   (D) Perform histogram equalization on the original images and transformed images.

   (E) Compare the effect of Contrasting stretching on the images Figure 1. Which of the images benefited best from contrast stretching and why ?

   (F) Convert the images to any 4 different color spaces. Comment on the histograms in the different color spaces with the histogram of the original RGB space.

   (G) What is the need for different color spaces? Comment on (one) advantage of the 4 different color spaces over RGB.



(a) (b) (c)

Figure 1

Section II - Kernels

(CO1,2&3)(22 marks)

1. Consider the image I given : (8 marks) $\begin{bmatrix} 50 & 130 & 10 & 50 & 70 \\ 130 & 240 & 130 & 40 & 10 \\ 240 & 130 & 20 & 50 & 42 \\ 50 & 240 & 50 & 20 & 50 \\ 50 & 13 & 20 & 70 & 72 \\ 42 & 245 & 52 & 245 & 132 \end{bmatrix}$

   (A) Write a function from scratch to convert a given image (8 bit) to 5 -bit and 3-bit quantization (use any method of your choice). Can images be converted from 5 bit to 8 bit?

   (B) Use box blur kernels (2D) (sizes : 2×2 and 3×3 ; sum over all pixels in each of the kernel equal to 24) to perform convolution and correlation on the 5 bit and 3 bit images without padding. Give the output matrices and compare the results.

   (C) Write a function from scratch to output Gaussian blur kernels (2D). The inputs to the functions : dimensions of the kernel and the variance σ.

   (D) Compute convolution and correlation using Gaussian blur kernel of dimension 3×3 with $\sigma = 3$ on the image I as well as its 5 bit and 3 bit counterparts.

2. Consider the grayscale images of the images (a,b,c) in the Figure 1. (14 marks)

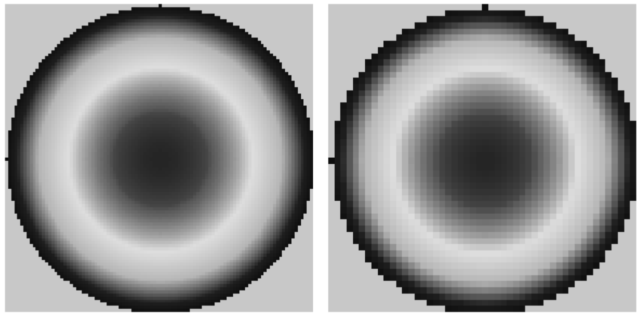
   (A) Add (a) Gaussian and (b) Salt and pepper noise to the grayscale images (Total 6 images). Compare the histograms to the original grayscale histogram.

   (B) (If necessary use padding to get same dimension outputs) Denoise using box kernels on the Images in steps (A), with parameters :

   - dimensions 3×3, 5×5 and 9×9
   - Sum over all the pixels in the kernel: $9, 24$ and 50.

   Note : total 9 box kernel :- dimension 3×3 with sum over all pixel $9, 24$ and 50, then similarly box kernel with dimension 5×5 with sum over all pixel $9, 24$ and 50, so on.

   (C) (If necessary use padding to get same dimension outputs) Denoise using Gaussian kernel on the Images in steps (A) , with parameters:



(a) (b)

Figure 2

- dimensions 3×3, 5×5 and 9×9
- $\sigma = 0.5, 4$ and 10.

Note : total 9 Gaussian kernels as in the case of (B) box kernel

(D) (If necessary use padding to get same dimension outputs) Denoise using median (dimensions $3 \times 3, 7 \times 7$ and 10×10) and Mean filters (dimensions $3 \times 3, 7 \times 7$ and 10×10) on the images in step A.

(E) (Table) Compare the performance of the the box kernels, gaussian kernel, mean and median filters by comparing the denoised images with the original image using metrics:

- Structural Similarity Index (SSIM)
- Mean Squared Error (MSE)
- Peak Signal-to-Noise Ratio (PSNR)

(F) (Comment on the results.) Rank (best-worst) the kernel/filters based on the metrics (SSIM,MSE and PSNR) for:-

- Gaussian noise
- Salt and pepper noise

Consider best of 2 from the 3 metrics (SSIM,MSE and PSNR for ranking) for ranking. For example if for an image $\hat{i}$, suppose MSE and SSIM score is better for kernel K_1 than for K_2, then K_1 rank greater than K_2, irrespective of the score of PSNR.

(G) Apply convolution using any 2 Laplacian kernels (dimension 3×3) on the outputs from steps B,C and D. What is the new ranking?

Section III - Fourier Transform
(CO1,2&3)(6.5 marks)

1. Consider the images in the Figure 2:-

(A) Both (a,b) are sampled at different rate from the same function. Which one has higher sampling rate and why?

(B) Plot the magnitude and phase of the Fourier transform (FFT) :

- Images (a,b) in Figure 2

- Images (a,b) after Spatial filtering using Bandpass
- Images (a,b) after Spatial filtering using Bandreject
- Images (a,b) after Rotating through angles $30°, -80°, 120°, -150°$ and $200°$.

References and Additional Instructions:

- Chapters 3,4 and 5 from the DIP book (referenced in the classroom) and Open-CV documentation. You can use inbuilt functions and LLMs. But the output and code should be there in the notebook submitted. You are free to attach a pdf file of the notebook.

- The viva will also include Lecture notes till the date of the deadline and questions related to work done in the assignment.

- If the comparison (SSIM,MSE and PSNR) is not in table format marks will be deducted.

Any code with run time error during demo (in any section I, II and III) will result in 0 marks. Do not use application such as Microsoft paint. Images generated without code will get -5 marks.

End